

CYBERSECURITY INTERNSHIP REPORT

INTRODUCTION

The report documents the operations completed during my cybersecurity internship, focused on network perimeter defense and traffic monitoring. The purpose of this project was to establish a secure localised network environment and evaluate communication protocol data.

The core objectives included threat analysis, host Firewall hardening, wireless gateway configuration and live packet inspection.

SUMMARY OF RESEARCHED NETWORK THREATS

- Worms: They are malware that is capable of self-replication, Needs no user intervention, Exploits Vulnerabilities to consume bandwidth. It can independently transmit itself to other systems on the same network.
- Virus: Malicious codes attaching to files. Requires human action to execute, It replicates across systems to corrupt data and systems, Viruses can spread through Software, Pirated Apps, USB Drives.
- Phishing: Social engineering Via deceptive communications. Tricks users into submitting private informations.
- Trojans: Destructive Software disguised as legitimate apps, opens hidden backdoor for unauthorized remote control.

NETWORK SECURITY AND HARDENING

OBJECTIVE

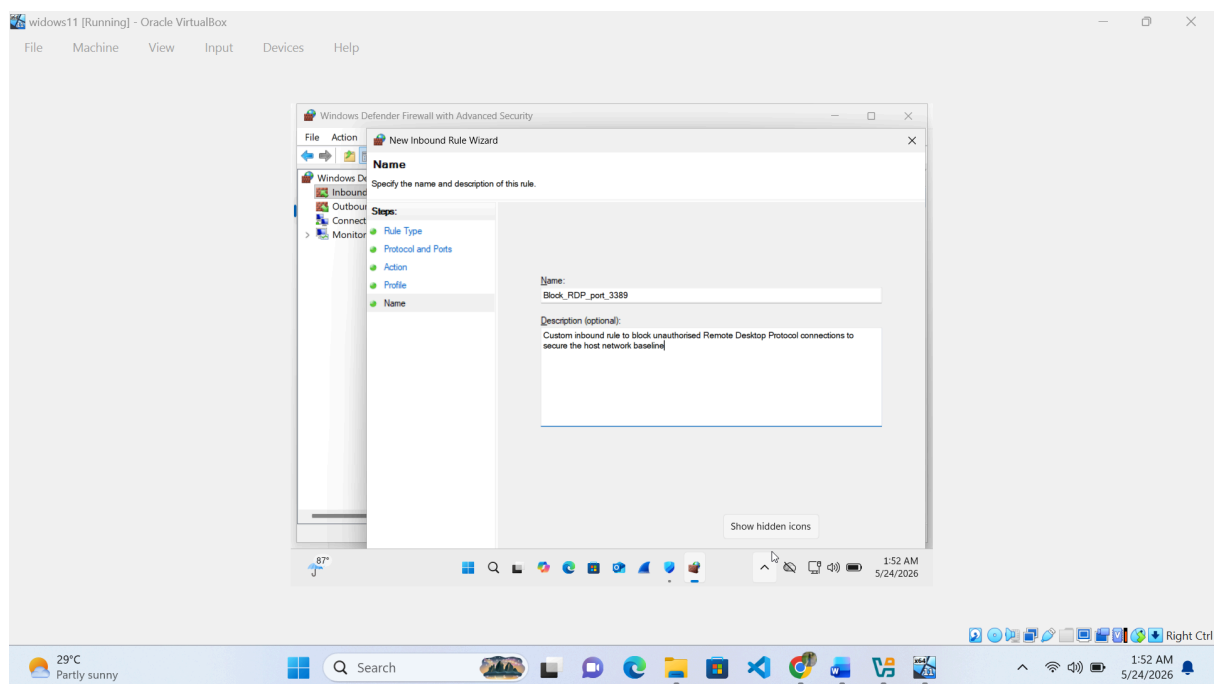
- To block vulnerable communication entry points on the host system.
- To evaluate the configuration of the home wireless router for safety gaps.
- To reduce overall vulnerability.

TOOLS

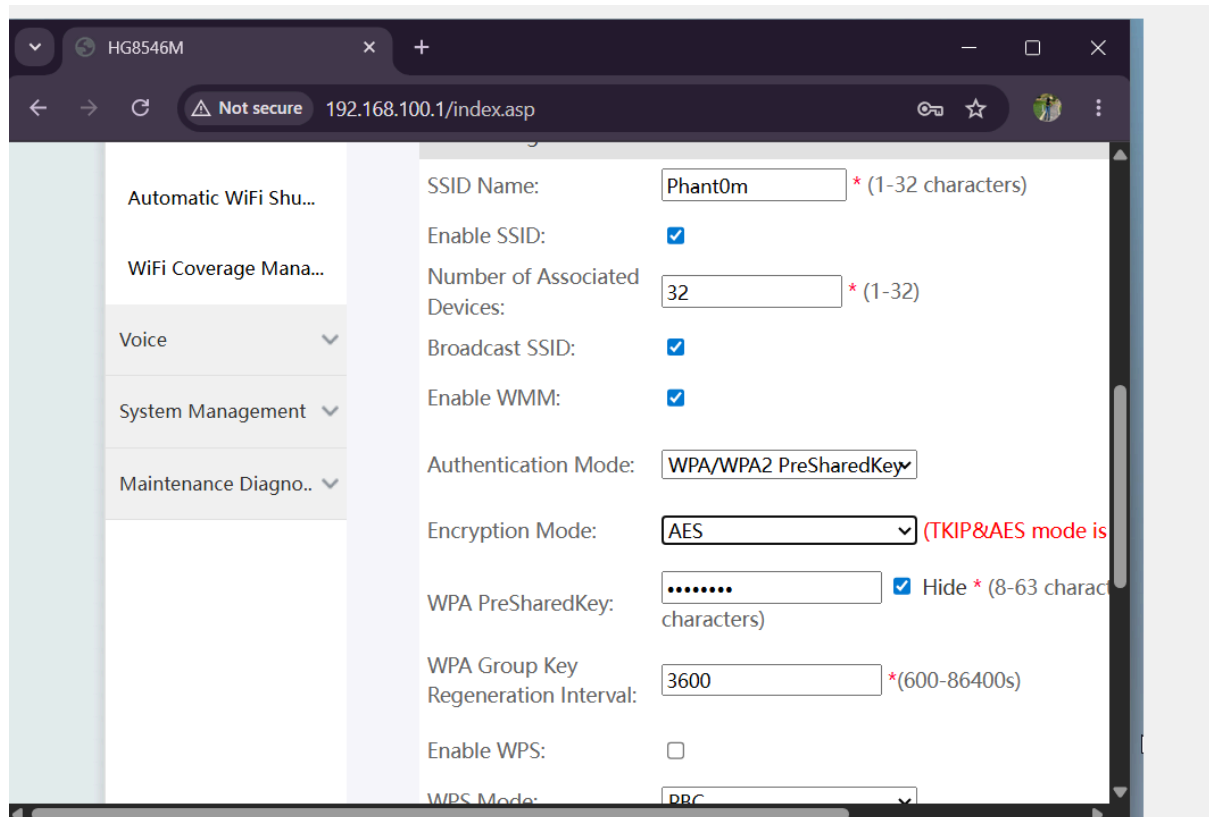
- Windows Defender Firewall: used advanced security to program custom inbound filtration policies.
- Huawei HG8546M Web Administration Interface: Layer 3 gateway portal used to evaluate wireless link parameters

STEPS TAKEN

1. Launched Advanced firewall on the Host Windows System to Monitor Current rule distributions.
2. Created a custom inbound rule targeting TCP port 3389 (Remote Desktop Protocol) to actively enforce traffic drops. (why RDP? Attackers frequently scan the internet for open ports to hijack computers remotely, Blocking it is a standard security best practice to prevent unauthorized access).



3. Logged in to the Router's Admin dashboard to check security profiles, encryption modes and active WPS parameters and made Sure WPS was disabled to prevent brute-force attacks.



4. Initiated Traffic Capture; Launched Wireshark on the Host System, Selected the active Ethernet interface, and Started A live Packet capture.

5. Generated Network Traffic: Opened Google Chrome and Navigated to neverssl.com to deliberately create plaintext HTTPS and DNS Network Communications.

(why neverssl.com? [Neverssl.com](https://neverssl.com) is a website that intentionally promises to never use encryption. It stays permanently unencrypted in plaintext HTTP, If an HTTPS packet is captured in Wireshark, it won't be readable.)

6. Analysed Protocol Packets: Returned to Wireshark and stopped the capture. Applied the `dns` filter to review domain resolution traffic on UDP port 53

7. Then applied the `http` filter to check for unencrypted web requests on TCP port 80

Below are Images for step 6 and 7 showing the captured packets.

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
6	0.072597600	10.0.2.15	192.168.100.1	DNS	79	Standard query 0x8388 HTTPS accou
7	0.073376300	10.0.2.15	192.168.100.1	DNS	79	Standard query 0xcb4d AAAA accoun
8	0.073944100	10.0.2.15	192.168.100.1	DNS	79	Standard query 0xdf6a A accounts.
9	0.086792900	192.168.100.1	10.0.2.15	DNS	129	Standard query response 0x8388 HT
10	0.086792900	192.168.100.1	10.0.2.15	DNS	107	Standard query response 0xcb4d AA
11	0.087285800	192.168.100.1	10.0.2.15	DNS	95	Standard query response 0xdf6a A
72	7.709622500	10.0.2.15	192.168.100.1	DNS	90	Standard query 0xbc0b A self.even
73	7.710265900	10.0.2.15	192.168.100.1	DNS	90	Standard query 0x8fd7 AAAA self.e
74	7.722378400	192.168.100.1	10.0.2.15	DNS	106	Standard query response 0xbc0b A
75	7.722378400	192.168.100.1	10.0.2.15	DNS	256	Standard query response 0x8fd7 AA
181	8.434688200	10.0.2.15	192.168.100.1	DNS	71	Standard query 0xefd9 HTTPS www.m
182	8.435179900	10.0.2.15	192.168.100.1	DNS	71	Standard query 0x7496 AAAA www.ms

> Frame 6: Packet, 79 bytes on wire (632 bits), 79 bytes captured
 > Ethernet II, Src: PCSSystemtec_9b:ce:c7 (08:00:27:9b:ce:c7), Dst:
 > Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.168.100.1
 > User Datagram Protocol, Src Port: 57249, Dst Port: 53
 > Domain Name System (query)

0000 52 55 0a 00 02 02 08 00 27 9b ce c7 0
 0010 00 41 1b 2b 00 00 80 11 00 00 0a 00 0
 0020 64 01 df a1 00 35 00 2d 30 f7 83 88 0
 0030 00 00 00 00 00 00 08 61 63 63 6f 75 6
 0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 4

Capturing from Ethernet

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http

No.	Time	Source	Destination	Protocol	Length	Info
7149	782.433685200	192.168.100.1	10.0.2.15	HTTP	1333	HTTP/1.1 200 OK (text/js)
7154	782.442134400	192.168.100.1	10.0.2.15	HTTP	511	HTTP/1.1 200 OK (text/js)
7156	782.550558900	10.0.2.15	192.168.100.1	HTTP	643	GET /images/loginbg.jpg HTTP/1.1
7158	782.554964700	192.168.100.1	10.0.2.15	HTTP	171	HTTP/1.1 304 Not Modified
7159	782.566326500	10.0.2.15	192.168.100.1	HTTP	597	GET /images/hwlog.png HTTP/1.1
7161	782.573496100	192.168.100.1	10.0.2.15	HTTP	800	HTTP/1.1 200 OK (text/html)
7166	782.578505300	10.0.2.15	192.168.100.1	HTTP	653	GET /images/greenline.jpg HTTP/1.1
7168	782.583017400	192.168.100.1	10.0.2.15	HTTP	171	HTTP/1.1 304 Not Modified
7169	782.602528400	10.0.2.15	192.168.100.1	HTTP	605	GET /images/copyright_log.png HTT
7172	782.607474500	192.168.100.1	10.0.2.15	HTTP	596	HTTP/1.1 200 OK (text/html)
7177	782.626603000	10.0.2.15	192.168.100.1	HTTP	549	GET /frameaspdes/english/ssmpdes.
7179	782.631028800	192.168.100.1	10.0.2.15	HTTP	171	HTTP/1.1 304 Not Modified

> Frame 1: Packet, 560 bytes on wire (4480 bits), 560 bytes captured
 > Ethernet II, Src: PCSSystemtec_9b:ce:c7 (08:00:27:9b:ce:c7), Dst:
 > Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.168.100.1
 > Transmission Control Protocol, Src Port: 53382, Dst Port: 80, Seq:
 > Hypertext Transfer Protocol

0000 52 55 0a 00 02 02 08 00 27 9b ce c7 0
 0010 02 22 1b 29 40 00 80 06 00 00 0a 00 0
 0020 64 01 d0 86 00 50 e5 0c fd 14 0d a8 3
 0030 ff ff 32 cd 00 00 47 45 54 20 2f 68 7
 0040 73 73 6d 70 2f 63 6f 6d 6d 6f 6e 2f 7
 0050 65 73 68 54 69 6d 65 2e 61 73 70 3f 5
 0060 37 39 36 31 33 35 36 31 39 30 30 20 4
 0070 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 3
 0080 31 36 38 2e 31 30 30 2e 31 0d 0a 43 6
 0090 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 6
 00a0 65 0d 0a 58 2d 52 65 71 75 65 73 74 6
 00b0 69 74 68 3a 20 58 4d 4c 48 74 74 70 5
 00c0 65 73 74 0d 0a 55 73 65 72 2d 41 67 6

Outlook

87° 2:26 AM

CHALLENGES

- Secure HTTPS websites encrypts traffic, making protocol fields unreadable. Overcome by using a dedicated testing website like neverssl.com to generate visible cleartext packets.
- Modifying Router settings could disrupt household network clients. Overcome by leaving the Router encryption on its dual TKIP/AES setting to maintain backward compatibility while documenting the risks.

RESULTS AND OUTCOMES

- Built a comprehensive understanding of automated and human-target malware threats.
- Successfully blocked inbound Remote Exploits targeting port 3389.
- Confirmed the Wireless network edge is secure against WPS pin cracking Utilities
- Isolated Cleartext DNS and HTTP packets, exposing how easily unencrypted traffic can be monitored.